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Invited Review

Automated sortation conveyors: A survey from an operational research perspective☆

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ABSTRACT

Global trends such as just-in-time, containerization, and e-commerce have led to an ever increasing freight volume to be transported under tight delivery schedules. Many supply chains, thus, apply fully-automated sorting systems as one basic component of their distribution processes. Examples range from baggage handling in airports, over parcels sorted by tilt tray conveyors in distribution centers of the postal service industry, up to batches of picking orders isolated in the warehouses of e-commerce retailers. This paper surveys the scientific literature on all kinds of fully-automated conveyor-based sorting systems from an operational research perspective. We describe the wide range of applications and their different sorting systems, define their joint decision problems to be solved when designing and operating a sorter, review the literature, and identify future research challenges.

1. Introduction

Nowadays, automated sorting systems (ASS) have become an essential component in many supply chains, mainly because of four elementary characteristics:

- ASS are fast: Next-day or even same-day deliveries are an elementary promise of many online retailers (Yaman, Karasan, & Kara, 2012) and postal service providers (Werners & Wülfing, 2010). When subtracting the transportation times this leaves only narrow time windows for all retrieval and sorting operations. Without automated sorting these targets seem barely realizable.
- ASS handle a huge amount of shipments: For instance, the air hub in Cologne (Germany), which is one of world-wide three main hubs in postal service provider UPS' express logistics network, has a capacity for sorting up to 190,000 shipments per hour (Airportzentrale, 2014).
- ASS are reliable: The U.S. Department of Transportation (2009), for instance, reports a range from 1.42 to 9.17 when ranking 19 U.S. airlines with regard to the mishandled baggage per 1.000 passengers. Clearly, any lost or delayed baggage causes loss in goodwill so that a reliable automated baggage handling became a critical success factor in the airline industry.
- ASS handle heavy and bulky items: The aging societies in many industrialized countries suggest to automate activities causing high ergonomic stress for human workers (Otto, Boysen, Scholl, & Walter, 2017). The repeated handling of heavy shipments is certainly among these activities so that ASS handling oversized shipments have a long tradition in postal distribution centers (Fedtke & Boysen, 2017b) and cross docks (Bartholdi & Gue, 2000).

Although the basic layout of an ASS is fairly simple – it mainly consists of one or multiple input stations for receiving, a network of conveyor segments, and multiple outbound stations for collecting the shipments sorted by addressee – quite a few decision problems need to be solved when designing and operating an ASS. This paper surveys ASS from an operational research (OR) perspective. We introduce the main fields of application for ASS and describe the basic strategic and operational decision problems. Furthermore, we survey the existing scientific literature and identify future research challenges.

For this purpose, the remainder of the paper is structured as follows. Section 2 defines the scope of this survey. Then, Sections 3–6 address the main fields of application for ASS. For

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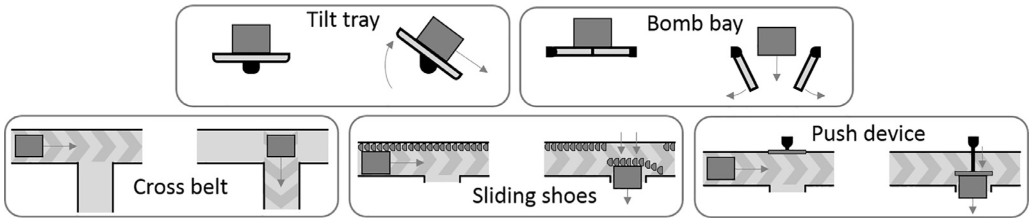

Fig. 1. Alternative sorter technologies.

each field, namely ASS in warehouses, cross docks, in the postal service industry, and in airports, we describe the layout of their typical ASS, define the basic decision problems to be solved, and review the literature. Finally, the results, key findings, and future research needs are discussed in Section 7.

2. Scope of survey

An ASS is a fully-automated conveyor system where an intermixed sequence of inbound shipments, which enter the system via one or multiple inbound stations, is sorted into outbound stations assigned to different addressees. According to our definition an ASS consists of three basic elements: (i) inbound stations, (ii) the main conveyor, and (iii) outbound stations. To realize a specific ASS implementation these elements can be differentiated as follows:

(i) In its most basic form an ASS has only a *single* inbound station so that all shipments enter the system via the same access point. A typical inbound station consists of a conveyor belt, first, moving shipments towards a recognition system, where they are weighed and identified, e.g., by scanning a bar code attached to a piece of baggage (Abdelghany, Abdelghany, & Narasimhan, 2006), by identifying a handwritten address on a parcel by some optical character recognition (OCR) software (Fedtke & Boysen, 2017b) or by receiving information from an RFID chip (Ilie-Zudor, Kemény, Van Blommestein, Monostori, & Van Der Meulen, 2011). Afterwards, a switch system successively loads the shipments onto the main conveyor and ensures an appropriate distance among each other. Alternatively, the main conveyor can also manually be loaded by human workers. Note that the succession of identification and loading can also be reverted. Larger ASS often have *multiple* inbound stations. For instance, the central airhub of postal service provider DHL in Leipzig (Germany) has four inbound stations (Fedtke & Boysen, 2017b). Multiple access points increase the infeed capacity and avoid a bottleneck on the loading side. Properly spaced and alternatingly placed with outbound stations, multiple inbound stations allow for multiple loads of the same conveyor segment per cycle. Each occupied position on the main conveyor can be refilled at the successive inbound station, if the shipment reaches its dedicated outbound station beforehand.

(ii) The main conveyor steadily moves loaded shipments along a fixed pathway, while passing junctions towards outbound stations. A typical velocity for such a conveyor is about 2.5 meters per second (Fedtke & Boysen, 2017b). There are different technical solutions of how a shipment is redirected into its outbound station (see Fig. 1). A wide-spread solution are tilt tray conveyors consisting of distinct trays each of which to be loaded with a single shipment. Each tray is tiltable to one or both sides so that a shipment can slide into its dedicated outbound station. Other solutions are bomb bay sorters, where the floor plate is opened and the shipment falls into a large bag or container, and cross belt sorters, where each conveyor segment is a small conveyor by itself operating at right angles towards the direction of the main conveyor. Finally, also some form of sliding shoes or other push mechanism can be applied. From an OR perspective, the technical realization of the sort mechanism is less of an issue. It is rather the structure of the pathway, which influences the operational processes. In its most basic form the pathway is just a *line*. A line requires that undelivered shipments, e.g., due to a tailback at an outbound station, are collected at the end, transported back to the access point, and reinserted into the system from time to time. This additional effort can be avoided if the pathway is a closed *loop* (resp. recirculation conveyor, loop sorter). Then, undelivered shipments can cycle on the loop until congestion is eliminated. However, there also exist *bi-directional* systems consisting of multiple loops arranged above each other (Briskorn, Emde, & Boysen, 2017). These systems allow to direct shipments into both directions so that the shorter path towards the respective outbound station can be selected. Finally, the pathway may consist of a complex *network of conveyors*. For instance, Jarrah, Qi, and Bard (2014) report on a distribution center of a postal service provider applying a hierarchical system, i.e., consisting of pre-sorter plus final sorters. In this way, a decomposition into smaller subsystems is enabled. We like to emphasize here that normally it is sufficient to capture the structure of the pathway (e.g., loop) rather than the exact shape (e.g., circle).

(iii) Each outbound station can be assigned a specific addressee to receive all dedicated shipments. The appearance of an outbound station heavily depends on the field of application as well as the size and weight of the shipments. Wide-spread solutions are dead-end conveyor segments, e.g., piers for collecting all baggage for a specific flight (Ascó, Atkin, & Burke, 2014) or telescope conveyors to be pulled directly into a truck trailer (Fedtke & Boysen, 2017b). On the other hand, bomb-bays or chutes can also direct a shipment directly into a large bag or small transport container. Again, from an OR perspective the technical realization does, typically, not influence the decision problems. An important issue, however, is the durability of the destination assignments. If outbound destinations are served again and again, e.g., by multiple trucks each day, then *fixed* destination assignments are typically applied. This allows the workers to learn the topology of the terminal. Fixed destinations are often applied in the postal service industry, where they are altered just once every couple of months (Boysen & Fliedner, 2010). Online retailers supply a huge variety of customers so that under these circumstances *variable* destination assignments need to be applied. Here, an outbound station is only briefly assigned to a specific order until all items are collected. Then, the station is reassigned to receive the next order.

By combining these characteristics of our three basic system elements different ASS can be realized. The schematic layout of a multi-inbound, loop-based tilt tray system, which is, for instance, often applied in the postal service industry, is depicted in the left part of Fig. 2. In the following, we briefly sketch the main fields of application where ASS are applied (see the right part of Fig. 2). We systemize these applications by the source of its inbound stream, which may either stem from a warehouse or has just been

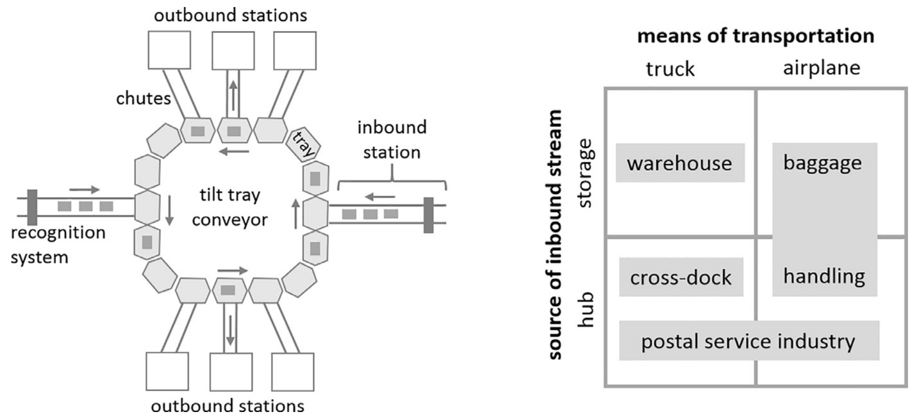

Fig. 2. Schematic ASS layout (left) and fields of application (right).

delivered in a hub environment, and the dedicated means of transportation, i.e., truck or aircraft.

- The typical applications of ASS in *Warehouses*, i.e., facilities to intermediately store goods in between two successive stages of a supply chain (see Gu, Goetschalckx, & McGinnis, 2007), are twofold. First, ASS play a crucial role if a batching and/or zoning policy is applied (see De Koster, Le-Duc, & Roodbergen, 2007). Batching means that multiple customer orders (i.e., a batch) are jointly collected from the shelves by a picker so that more efficient picker tours through the warehouse are enabled. Under a zoning policy, the warehouse is subdivided into multiple areas from which order pickers pick the part of the order that is in his/her assigned zone. In both cases, an ASS can be applied to split the collected items and to accumulate the customer orders. Once each picking order is readily packed into a dedicated shipment, ASS can also be applied to sort these packages by their dedicated shipping ways. The outbound stations of these ASS are often telescope conveyors to directly transport the parcels into the truck trailers of different transport providers.
- *Cross docks* are special distribution terminals without long-term storage dedicated to a consolidation of smaller shipments to full truckloads (see Boysen & Fliedner, 2010; Buijs, Vis, & Carlo, 2014; Van Belle, Valckenaers, & Cattrysse, 2012). These terminals often apply closed-loop sub-surface chain conveyors (also denoted as draglines, see Bartholdi & Gue, 2000). In these systems, hand pallet trucks carrying pallets of goods can be coupled with the chain, which pulls them through the terminal. This way, pallets pass by their dedicated trailers so that logistic workers can release the respective pallet truck and can directly move the pallet into the trailer. Alternatively, the literature also reports on less-than truckload logistics providers applying belt-based ASS to consolidate shipments (see Boysen & Fliedner, 2010).
- In the *postal service industry*, sorting mail and parcels is barely thinkable without ASS. Postal service providers, such as UPS and DHL, apply huge systems consisting of complex conveyor networks, e.g. (McWilliams, Stanfield, & Geiger, 2005), and bi-directional loops with multiple inbound stations as described in (Fedtke & Boysen, 2017b). They exist for both truck-based continental transports Fedtke & Boysen, 2017b and airborne intercontinental services (Boysen & Fliedner, 2011).
- *Baggage handling* is among the most elementary services in any airport. Inbound baggage, i.e., newly arriving with passengers from the air- or landside or early check-in baggage from a storage area, needs to be delivered towards baggage sorting stations each assigned with an outgoing flight and terminating baggage of arriving flights needs to be transported towards baggage claims (Tošić, 1992). The huge amount of baggage to be handled in modern airports and the tight schedules make ASS an important facility in any airport.

Note that a more detailed description of the ASS applied in each field is given in Sections 3–6 each dedicated to one of these fields of application. Our (positive) definition of the paper's scope is now amended by a (negative) demarcation from related fields. We restrict our survey on conveyor-based systems where the path of a shipment is fixed for a given pair of inbound and outbound station. On this path, shipments do not move individually; that is at each moment all shipments on the conveyor move with the same speed.

Obviously, sorting can also be achieved by moving the shipments more individually, for example by placing each shipment on separate vehicles, which can either individually travel along a shared pathway or can reach their dedicated outbound stations on facultative paths. In this case, however, the routing aspect, i.e., the position of shipments over time, often is predominant. Corresponding optimization problems, therefore, are rather close to vehicle routing problems and pick-up and delivery problems, which are very well covered by literature surveys, e.g., Laporte (1992) and Berbeglia, Cordeau, Gribkovskaia, and Laporte (2007). Our restriction excludes the following (related) sorting and transportation systems.

- In container ports, automated guided vehicles (AGVs) transport containers between quay cranes servicing vessels and yard cranes of multiple container blocks (Vis, 2006). AGVs can travel on individual paths (on a possibly restricted network) and are, thus, excluded. AGVs are also applied in warehouses to preposition items, e.g., in the KIVA system, see (Boysen, Briskorn, & Emde, 2017a). These systems are also excluded.
- Individual vehicles moving along ever changing paths are also applied in an innovative sorting system based on autonomous robots (see Tompkins International, 2017; People's daily, 2017). Each robot of a large fleet is loaded with an item and moves along a grid of squares on the shop floor towards one of many scuttles dedicated to the item's destination. Here, the robot tilts its tray so that the items on top slides into the scuttle where items are collected.
- For baggage handling, there exist systems where each piece of baggage is placed on an individual destination-coded vehicle (see Tarău, De Schutter, & Hellendoorn, 2011). These unmanned carts propelled by linear induction motors are mounted on tracks, but can travel with varying velocity and distance among each other. Similar systems are also applied for supplying parts in production environments (see Le-Anh & De Koster, 2005).

- Loop-based automated handling systems are applied in production environments to deliver workpieces towards assembly stations. Nazzal and El-Nashar (2007), for instance, survey these systems in wafer production systems; other hoist-based systems are applied in the electronics industry (see Manier & Bloch, 2003 and Leung, Zhang, Yang, Mak, & Lam, 2004). However, these systems typically have to successively visit multiple stations for assembly operations, such that no direct relation to sorting exists.
- Recent prototypes try to apply the basic logic of ASS for even larger shipments such as containers. They are, for instance, applied in railway yards where gantry cranes consolidate containers among different freight trains (see Boysen, Fliedner, Jaehn, & Pesch, 2013b). To relieve the cranes from long horizontal travel along the yard (and the crane interference such a movement would induce) modern yards apply rail-bound shuttle cars, which move containers between crane areas (see Boysen & Fliedner, 2010). These shuttles, however, move individually along the tracks so that these systems are excluded.
- Another sorting system in the rail context are traditional shunting yards. Here, inbound waggons are uncoupled, moved over a shunting hill, and – when rolling down – directed onto outbound tracks via switches. This way, intermixed inbound trains are assembled (sorted) to outbound trains heading towards their respective destination. A recent survey on shunting yards is provided by Boysen, Fliedner, Jaehn, and Pesch (2012). However, the movement downwards the shunting hill can facultatively be retarded with the help of break shoes so that no steady movement arises.

Our survey is not dedicated to technical aspects, such as the applied hardware, but rather takes the OR perspective. We, thus, focus the main decision problems to be solved when planning and designing a new ASS (or altering an existing one) and when operating an ASS. Ordered with regard to their planning horizon, we structure the multitude of decision problems into the following four areas:

- *Layout* problems need to be solved if a new system is designed or an existing one is to be altered. In these cases, layout alternatives need to be evaluated according to the basic trade-off between performance and investment costs. Specifically, the decision problems of this phase have to decide on all structural aspects, such as the number and locations of inbound stations, structure and length of the main conveyor (e.g., single-loop vs. bi-directional system), and the number and locations of outbound stations.
- On the *inbound side*, the way of the shipments into an ASS has to be structured. This includes the choice of the inbound station for each shipment and the feeding sequence per station. In many terminals of the postal service industry, for instance, inbound trucks can be unloaded at different sides of the terminal so that alternative inbound stations are applied. This way, an appropriate assignment of unloading docks to trucks can influence the sortation performance (see Boysen, Fedtke, & Weidinger, 2017b).
- On the *outbound side*, destinations (addressees) have to be assigned to outbound stations during ASS operations. In the literature, this problem is also denoted as the destination assignment problem (Boysen & Fliedner, 2010). The structure of this problem is mainly affected depending on whether a fixed (e.g., valid for several months) or a variable assignment that continuously alters is sought. Other decision problems on the outbound side are related to the staffing of the packing operations, which are required once shipments have reached their outbound stations.
- Finally, short-term (or even real-time) *scheduling* decisions have to be made. They are, for instance, related to the point in time a shipment should be loaded onto the main conveyor. If alternatives are available, the specific path (in the network of conveyors) for each shipment has to be determined (see Briskorn et al., 2017).

The multitude of ASS applications make it anything but astounding that already some related survey papers exist. For instance, there are the surveys of De Koster et al. (2007) on warehousing, of Tošić (1992) and Ashford, Coutu, and Beasley (2013) on airport operations, and of Boysen and Fliedner (2010), Van Belle et al. (2012), Buijs et al. (2014), and Ladier and Alpan (2016) on cross docks. They all mention ASS as one aspect in a much broader context. Only the survey paper of Muth and White (1979) is explicitly dealing with sorting systems. On the one hand, this paper is already quite old so that plenty additional papers have accumulated in the meantime. Moreover, the focus of their survey is on analytical methods specifically dedicated to layout aspects. Thus, an up-to-date survey paper on ASS seems well justified.

Finally, we briefly specify our database search for retrieving the papers to be reviewed (see, e.g., Hochrein & Glock, 2012 for a general description of how to set up a systematic literature retrieval). As keywords specifying our domain we apply “sorter”, “sorting system”, “sortation system”, “sortation conveyor, “automated sorting”, and “order accumulation system”. Furthermore, a second group of keywords, i.e., “warehousing”, “warehouse”, “cross dock”, “cross docking”, “postal service”, and “baggage handling” is applied to specify the field of application. Any combination of keywords from first and second group has been applied as a query in two scholarly databases, namely Business Source Premier and Scopus. All English-language papers published in peer-reviewed journals that have been retrieved and those cited in their reference lists (snowball approach) were checked for relevance by analyzing their abstracts. Additionally, some selected working papers and conference proceedings have been integrated, if (according to the authors’ subjective assessment) they considerably contribute to the surveyed field.

3. ASS in warehouses

Warehouses (also denoted as distribution or fulfilment centers) are applied to intermediately store goods in between two successive stages of a supply chain (Gu et al., 2007). In the most basic setting, each single order demanding some stored items is individually collected from the shelves so that no ASS is required. In the e-commerce era, however, next- or even same-day deliveries promised to the customers put increasing stress on warehouse operations (Yaman et al., 2012). Thus, to enable more efficient warehousing processes, ASS are typically applied in two different situations:

- To enable a more efficient picker-to-parts order picking process many warehouses apply batching and/or zoning strategies (see De Koster et al., 2007). The former means that multiple orders are unified to a batch (or wave) of orders jointly assembled on a picker tour so that the pick density per tour is increased. Applying a zoning policy means to partition the warehouse into smaller areas so that order pickers only pick the part of an order that is stored in their assigned zone. In this way, parallel order picking is enabled and each picker only traverses smaller areas of the warehouse. Both policies lead to intermixed items, which need to be separated into customer orders. Some warehouses apply manual sorting processes via so-called put-walls (Boysen, Stephan, & Weidinger, 2018), but mostly ASS are applied like they are depicted in Fig. 3 (middle). These ASS channel items into outbound stations called accumulation or

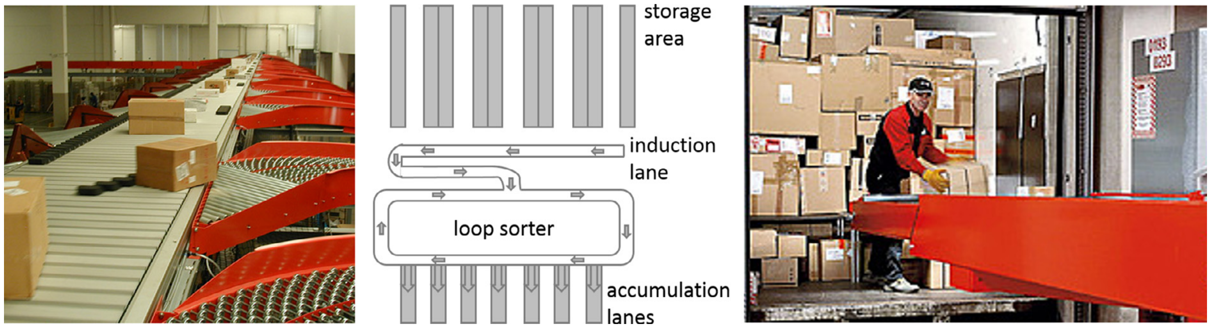

Fig. 3. Sliding shoe sorter POSISORTER™ (Source: Vanderlande) (left), a typical sorter setting in a warehouse (middle) and telescope conveyor (Source: Budde Fördertechnik) (right).

packing lanes where completed orders are packed into cardboard boxes and forwarded to the shipping area.

- In the shipping area, many warehouses apply another ASS to direct the stream of parcels towards different outbound trailers. These trailers are dedicated to different outbound destinations or belong to different transport providers such as DHL and UPS (Johnson, 1998). In this application, outbound stations are typically telescope conveyors (see Fig. 3 (right), which can be extended to directly lead into the trailers (Boysen et al., 2017b).

Many warehouses apply sliding shoe sorters like they are depicted in Fig. 3 (left), but also tilt tray, cross belt, and bomb bay sorters can be found. The most wide-spread ASS layout is depicted in Fig. 3 (middle). There is only a *single* inbound station injecting either the picked items or the packed customers parcels onto the main sorter. Typically, a *loop* sorter is applied (Johnson & Lofgren, 1994; Meller, 1997), but also *line* sorters can be found (Boysen, Fedtke, & Weidinger, 2016; Owyong & Yih, 2006). Bi-directional or even more complicated networks, however, are rather uncommon. On the outbound side, we have the accumulation lanes which either collect individual picking orders or shipments for distinct transport providers. In the former case, the vast number of potential customers requires that the assignment of outbound destinations (customers) to outbound stations (accumulation lanes) is *variable*, whereas the subcontracted transport providers often remain stable over a longer period of time so that also a *fixed* assignment is possible. The following three sections review the literature related to the main decision problems arising when applying an ASS in a warehouse.

3.1. Layout planning

During the design phase, the physical layout of an ASS has to be determined. The main system elements that need to be selected during this phase and whose specification can be supported by OR methods are the following:

- selection of either line and loop sorter,
- capacity (length) of the sorter,
- velocity of the sorter, and
- number and capacity of accumulation lanes.

To quickly evaluate the trade-off between investment costs and operational performance of different elements and layouts, mainly fast methods such as analytical methods and simulation studies are suggested by the existing literature.

Bozer and Sharp (1989) evaluate different sorter layouts for warehouse sortation systems with wave picking. In the simulated generic system, waves are initially released to a single induction lane. The items may then leave the system through one of multiple accumulation lanes. Items that do not leave the sorter circulate on a single recirculation conveyor. Finally, items leaving the system through the accumulation lanes are handled by a worker, who removes items from the lanes with a deterministic rate. In a slightly alternative setting, which is also simulated, the sorter does not allow recirculations at all, while the remaining setup remains identical. In that case, items are held at the end of the induction lane, if none of the suitable accumulation lanes is free. To account for stochasticity of the picking process, the authors assume that items entering the induction lane follow a Poisson distribution, while the destinations of the items are uniformly distributed. Service rates of operations on all lanes are deterministic, the system's total input capacity equals its total output capacity, and recirculation is only applied if an accumulation lane is full, i.e., failures like reading label errors are not considered. The simulation results show that in a system without recirculation the throughput ratio of the system decreases if the number of accumulation lanes increases. The throughput of a system with recirculation depends on the number of accumulation lanes, if those are uncapacitated. However, if the accumulation lanes are capacitated, the throughput capacity increases with the number of accumulation lanes and the capacity of the induction lane.

Johnson and Lofgren (1994) model the design and processes of Hewlett-Packard's new North American distribution center, which is designed as follows: In a hardware storage area ordered items are picked up and brought either to a bulk-shipping area, the repackaging area or the picking area. Items arriving at the bulk-shipping area are manually brought to the right pallet and are then directly loaded onto trucks. Items arriving in the repackaging area are repacked and smaller by-items, like manuals, are added, before they are forwarded to the recirculation conveyor (i.e., after quality assurance as well as filling and taping). Items arriving at the picking area are bundled and passed on to a single induction lane, leading into the recirculation conveyor in waves. Those waves are subsequently queued and only allowed to enter the recirculation conveyor whenever one of the multiple accumulation lanes, leaving the conveyor of the sortation system, gets available. The wave release rules in this pull process, but also the conveyor belt speed, the processing rates along the lanes, the size of the conveyor loop, and the lengths of the accumulation lanes are varied in the simulation study. Note that the sizes of conveyor loop and accumulation lanes affect the physical layout. The results of the study indicate that overlong conveyor loops result in deserted accumulation lanes, while longer accumulation lanes cause a decrease in the number of recirculations. Furthermore, releasing the waves too quickly often leads to overfull accumulation lanes, while delaying the release of a wave too long leads to deserted accumulation lanes.

On an even more strategic level, Russell and Meller (2003) and Petersen (2000) investigate the question, which warehouse environments justify the application of ASS at all. Petersen (2000) evaluates several order picking strategies that result from different warehouse settings, which differ in their degree of automation. He compares several policies, namely, strict order picking, batch picking, sequential zone picking, batch zone picking, and wave picking. The results show, that especially those strategies that require some form of ASS for a later order consolidation (i.e., batch and wave picking) prove to be the most efficient ones. However, Petersen (2000) only exploits the benefit of picking policies but neglects cost aspects for the required sorter. This point is treated by Russell and Meller (2003), who investigate the benefit of ASS compared to manual sorting processes with regard to several system parameters. The authors propose a descriptive model that includes demand and labor rates, fixed and variable costs for the sorting process, order and wave sizes, as well as the sorter capacity (i.e., the number of employees, packing stations, and induction stations required to meet demand). A prescriptive cost based optimization model compares the total annual system cost for several parameter configurations and recommends the system configuration that meets the given throughput. A sensitivity analysis shows, that varying a single parameter may not change the recommended system configuration, but varying several parameters simultaneously has a significant effect on the system. For high demand rates and low automated pack station cost, they recommend ASS, while at lower costs or lower demand manual sorting shows preferable. The authors furthermore conduct a simulation study, including stochastic aspects, to verify their results and provide an approximate analytical model to determine the optimal batching level in manual sorting systems, as it is identified as one of the most critical decisions.

3.2. Inbound problems

Typically, ASS in warehouses have just a single inbound station so that the composition of the inbound stream fed into the sorter is the main lever on the inbound side. When sorting customer orders, there are typically far more orders than accumulation lanes available. Whenever all these lanes are still occupied, newly arriving orders cannot be assigned a lane and, in the worst case, accumulating upstream congestion (also called gridlock) necessitates costly and laborious recovery procedures of logistics workers (see Gallien & Weber, 2010). This risk can be avoided if a wave-picking approach is applied. If the number of orders forming a wave/batch does not exceed the number of accumulation lanes, then waves just have to be released successively, such that all packing lanes are unoccupied before the next wave arrives. However, such a sequential policy leads to a low utilization of the sorter and an inefficient picking process in the storage area due to an unavoidable imbalance of the pickers' workload (Petersen, 2000). To avoid this drawback waves are often released in an overlapping manner. Once a specific completion threshold is reached, e.g., Bozer, Quiroz, and Sharp (1988) suggest 90% and Johnson and Lofgren (1994) report of 50% of all orders, the next wave is released. This, again, bears the risk of blockings if waves are released too early. Thus, existing literature on inbound problems of ASS in warehouses mainly investigates the delicate steering between gridlocks and efficient capacity utilization.

Gallien and Weber (2010) develop a model for an automated warehouse of a leading US online retailer, which considers different policies for releasing waves of bundled items into the main conveyor system in order to accumulate customer orders. This includes a classical non-overlapping wave release policy, in which the picking process of a batch of orders can only start once the previous order batch was released to the main conveyor system. Furthermore, they investigate possible improvements by applying alternative release rules allowing overlapping waves, which means that the picking of one order batch can already start while the previous one is still waiting for release. Finally, they compare the results with a split sorter policy, in which pickers can pack a new batch of orders in one half of the inbound system, while the other half of the inbound system contains a completed wave waiting for release. In their work, the authors develop and validate a model for waveless policies that predicts congestion and use it to derive operational control guidelines, that maximize the throughput but keep risk of gridlocks under a given threshold. The results of the study indicate that a waveless policy clearly outperforms all of the wave-based release policies in terms of throughput. Furthermore, the probability of any stagnation in a sortation system with a waveless policy is lower than in all wave-based policy driven sortation systems.

The effects of order-picking operations on order-consolidation time in a module-based fulfillment center are investigated by Owyong and Yih (2006). Such systems consist of several pick modules commonly stacked together, whereby each module contains two rows of carton flow racks and a takeaway belt. The application of split-case or parallel batch picking policies often leads to congestion in the packaging department. Their objective is to resolve such problems by reducing the estimated order-consolidation time. Although, the ASS is not explicitly modeled, the study is based on a system for automated mass sortation with a multiple induction station, which receives input from the takeaway belts. The authors develop an algorithm that alters the pick lists in order to locate items belonging to the same order in nearby bins. A simulation study shows, that the new algorithm outperforms the basic pick list generation process in the means of average order consolidation time.

A similar objective is pursued by Boysen et al. (2016), although the warehouse setup differs significantly. Here, bins with customer orders collected under a batching and zoning policy are temporarily stored in an automated storage and retrieval system (ASRS) before being released to the consolidation area. In a single induction station, a worker unloads the bins by placing the contained items on a linear crossbelt sorter that directs them to one of several packing stations. The authors introduce a surrogate objective, namely minimizing the order spread, that can be achieved by optimizing the bin release sequence from the ASRS. They formulate a mathematical model and propose and validate priority-rule-based and dynamic-programming-based solution procedures. With the help of a comprehensive simulation study, the appropriateness of the objective is validated. Observations suggest that optimized sequences prevent congestion at the packing station as long as they are the bottleneck of the consolidation process. Furthermore, the relationship between order picking and order sorting operations is investigated.

Kizilaslan, Bayraktar, and Eldemir (2103) describe an analytical model to integratedly consider wave picking and sortation operations. The goal is to calculate the wave size maximizing the overall efficiency. Here, a fundamental trade-off has to be taken into account. Increasing the wave size improves efficiency of the picking process while it increases processing times of orders during the sortation process (potentially due to congestions). The investigated layout is kept very simple and is split into the two process areas of interest: The picking area and the sortation area. Within the picking area items are picked from the shelves and put on a single takeaway conveyor belt. This conveyor belt transports the items to the sortation area, where it acts as the single induction lane to a recirculation conveyor. Finally, the readily assorted orders leave the recirculation. The authors suggest an analytical model that simultaneously includes picking and sorting operations and apply it to a case study.

3.3. Outbound problems and short-term scheduling

On the outbound side, accumulation lanes need to be assigned to outbound destinations. This is a challenging decision problem, especially if there are far more outbound destinations (customer orders) than lanes available. In this case, a short-term scheduling problem deciding on the assignment of lanes to any new customer order is to be solved.

Bozer et al. (1988) develop a SLAM simulation model to evaluate four different types of so-called wave profiles on an automated sortation system. The authors consider the most straightforward layout, consisting of a single induction lane, a recirculation conveyor and multiple accumulation lanes. All other operations, before or after reaching that part of the warehouse, are not considered. The wave profiles specify the number of customer orders accumulated in a single wave. For all four types of wave profiles, the authors apply four different distribution logics and compare their performance. The most intuitive logic, called incidental assignment, scans the system for an unassigned accumulation lane whenever an order is completely circulating within the loop sorter. As soon as such an unassigned accumulation lane is found, the order is assigned to that free lane once it passes the concerned lane. The biggest challenge with that approach is that an order might circulate indefinitely, if it does not pass the right accumulation lane in the right moment, as other orders passing by will then be selected instead. Consequently, the other three presented logics address this issue by storing all available orders in a queue and assigning the oldest, the largest or the smallest order to the next free accumulation lane. A simulation study figures out that the advantages of the incidental assignment strategy decrease, if the number of accumulation lanes is equal or greater than the average number of orders per wave.

Meller (1997) proposes an algorithm that assigns orders to accumulation lanes, while the arrival sequence of single items at the induction point is given. The objective of the algorithm is the minimization of the maximum time an order recirculates on the loop conveyor. The sortation system of a leading US American clothing manufacturer is used as a case study to test the algorithm. Again, the tested layout consists of a single induction lane, a recirculation conveyor and several accumulation lanes. Wave picking is applied where the size of a wave depends on a given number of trucks waiting for the corresponding orders. The lane assignment problem is represented as a mathematical model, which considers the assignment of several orders related to the same truck to one or more accumulation lanes. That problem is solved with a decomposition approach, solving minimum cardinality subproblems.

Johnson (1998) develops an analytical model of the sorting process on a generic sortation system, again, consisting of a single induction lane, a recirculation conveyor and multiple accumulation lanes. Wave picking is applied. The model proves that the strategy of assigning the next available order to an unassigned accumulation lane outperforms any fixed lane assignment rule, in terms of throughput and operation times, in systems with less or no blocking at the accumulation lanes.

4. ASS in cross docking terminals

A cross docking terminal is a special facility in a distribution network without long-term storage where smaller shipments are merged to full truck loads (Boysen & Fliedner, 2010). In these terminals, inbound trucks are docked at inbound gates, shipments are unloaded, sorted according to addressee, moved across the terminal and directly loaded into outbound trucks dedicated to the respective destinations. In this way, truck load factors are increased and economies in scale of transportation can be realized. Many cross docking facilities abstain from ASS. Here, sorting is manually executed by logistics workers and for moving shipments across the terminal forklifts or pallet trucks are utilized. However, there also exist terminals automating these processes with an ASS. Rarely, conveyor-based ASS (see Fig. 3 (left)) exist (Stephan & Boysen, 2011a). More often, in-floor chain conveyors systems (see Fig. 4 (left)) are applied, which are also denoted as draglines (Bartholdi & Gue, 2000) or tow lines (Bakst, Hoffner, & Jacoby, 1996; Pan, Ding, & Ford, 2008). The loop-shaped chain steadily moving inside the floor covering is accessible via a slit so that pallet trucks can be hooked. Next to the staging area of an inbound truck a pallet along with its pallet truck can be hooked by some logistic workers so that the load moves through the terminal and is hooked off once it passes its designated outbound dock. In some terminals, the identification of passing pallets to be loaded into a specific outbound trailer is supported by light signals. Fig. 4 (right) depicts the schematic layout of a cross docking terminal applying an in-floor chain conveyor.

As the chain is accessible next to each staging area, a typical ASS of a cross docking terminal has *multiple* inbound stations and the main conveyor is *loop*-shaped. Depending on the number of outbound destinations and their continuous delivery over time the assignment of outbound destinations to outbound stations can either be *variable* or *fixed*.

Although cross docking has received considerable attention in the scientific literature we hardly found any papers treating the ASS of cross docks in detail. Instead, the time required by the sorting process is merely modeled as a constant time span, which may vary between different inbound and outbound stations. Buffa (1986) shows as early as 1986 that logistics costs could always be reduced by integrating the simultaneous planning of inbound and outbound trucks into the management of a distribution system. Therefore, there seems to be almost no literature focusing on either inbound trucks or outbound trucks but typically both problems are considered in an integrated manner. Short-term scheduling of freight through an ASS within a cross dock is not considered in the literature either. Consequently, after considering layout planning in Section 4.1, we treat inbound and outbound problems in a single Section 4.2. We keep these two sections rather short since, as mentioned above, often the focus is not on sortation although it is implicitly considered.

4.1. Layout planning

A first fundamental question to be answered during a terminal's layout phase is whether an ASS is indeed required. Alternatively, no ASS is applied and shipments are moved by forklifts or pallet trucks (Vis & Roodbergen, 2008). To answer this question investment cost for the ASS need to be traded off against the higher operational costs of a non-automated movement of goods. Once a decision for an ASS is made, further layout aspects, such as the conveyor technology (in-floor chain conveyor vs. a belt system) as well as the path and capacity of the ASS have to be determined. A further layout aspect of the terminal, which also impacts ASS performance, is the question whether one side of the terminal is exclusively reserved for inbound operations and the other for outbound operations. Alternatively, inbound and outbound docks can arbitrarily be mixed along the perimeter of the building (see Stephan & Boysen, 2011b). The former has the advantage of a clear flow of shipments inside the terminal, but shipments loaded onto the ASS always move past all other inbound docks and the capacity is only freed in the outbound area. With an intermixed sequence of inbound and outbound docks multiple loads of the ASS at the same conveyor position are enabled per cycle of the conveyor. This relieves a bottleneck ASS in times when plenty shipments need to be moved concurrently. In spite of the plenty layout aspects to be investigated, literature in this field is rather scarce.

Rohrer (1995) focuses rather on the evaluation of specific layouts than on approaches to find good layouts. He stresses the importance of considering as many operational details as possible to guarantee general acceptance and applicability of the resulting

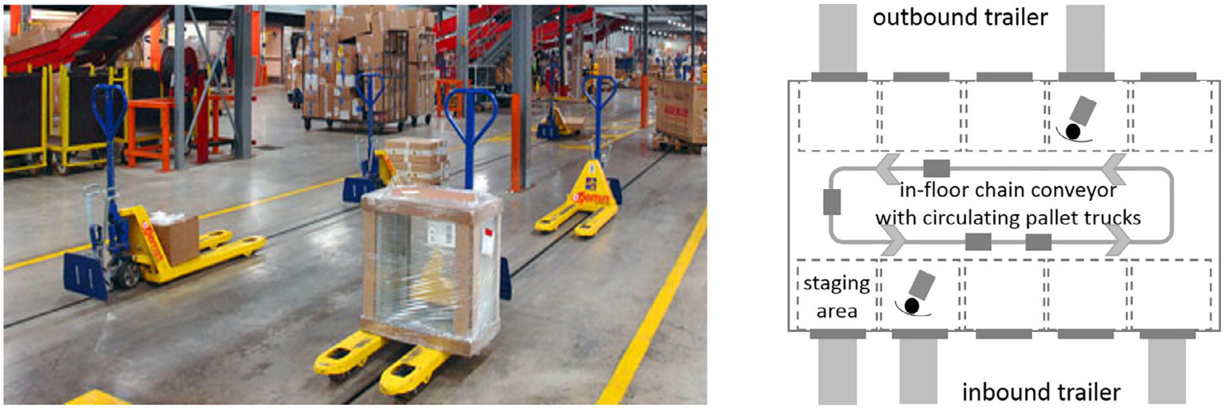

Fig. 4. In-floor chain conveyor system E'tow® (Source: Egemin Automation) (left) and a schematic layout of a cross docking terminal with sorter (right).

simulation model. This includes the hardware setup, like the use of bar code scanner or automated guided vehicles, as well as the software setup, which must include several clear operation rules that steer the overall system. Depending on the application, the author proposes to alter the output of the simulation accordingly and measure different system parameters, e.g. the performance of specific system parts, the conveyor usage rate or the truck handling time. Sharp and Liu (1990) explore the effects of shortcuts and offline spurs, i.e., separated tracks accessed via junctions, within a fixed-path closed-loop material system. By introducing an analytical model and tackling it with a general purpose integer linear programming algorithm, they determine good network configurations with respect to load-carrier and spur costs. A similar approach is chosen by Bartholdi and Gue (2000). A cost-minimizing steady-state model is formulated, which estimates the average waiting time to transfer all dragline freight from inbound to outbound trailers, including possible congestion of the system. They identify efficient layouts with the help of a simulated annealing procedure (SA) that interchanges pairs of trailers. Design aspects are also investigated by Al-Sultan and Bozer (1998) by simultaneously considering the path and layout configurations of a fixed-path material handling system. Promising layouts are identified by applying SA. Pan et al. (2008) formulate a multi-commodity maximum network flow model to determine the flow capacity of towline conveyor systems and optimize the workforce allocation in the system. Hereby, they aim to ease further capacity analysis and staffing allocation. The towline system of a publishing house that receives 500–1200 pallets and ships 350,000–600,000 books daily is investigated by Bakst et al. (1996). They apply a simulation model and conduct an analysis of the results. Van Belle et al. (2009) examine the concepts and principles of new-generation manufacturing execution systems and how they can be used to control the product flow. The resulting control system is applied to a cross docking distribution center with chain conveyors.

4.2. Inbound and outbound problems

The manifold literature on a planning of inbound and outbound flows in cross docking terminals can be separated into two streams of literature, which either aim at a *fixed* or *variable* assignment of destinations to dock doors (see also Boysen & Fliedner, 2010).

(a) In the former case, a fixed assignment remains stable over a longer period of time so that the trucks servicing respective in- and outbound destinations are always processed at same dock doors. The resulting assignment problems, typically, abstract from timing aspects (i.e., when exactly trucks arrive and depart) and aim to assign inbound destination that provide plenty shipments for specific outbound destinations are located close together. In this way, the distances to be bridged inside the terminal, e.g., on the dragline, are reduced so that transport resources are relieved and transshipment processes are accelerated.

(b) Especially if there are much more destinations than dock doors, the assignment is variable and each trailer receives an arbitrary available dock. This short-term view requires the integration of timing aspects so that truck assignments do not timely overlap at the same door. If the loads inside the trailers are known in advance, again, the movement of shipments inside the terminal can be reduced by a proper assignment of trailers and dock doors.

On both planning levels, ASS are not explicitly modeled in the existing literature. Consequently, we do not aim at an exhaustive survey on (a) the mid-term destination assignment problem and (b) the short-term truck scheduling problem. This task is already fulfilled by the in-depth review papers on cross docking (see Boysen & Fliedner, 2010; Buijs et al., 2014; Ladier & Alpan, 2016; Van Belle et al., 2012). We, however, review some relevant representatives for each case.

The problem of assigning doors to destinations over a longer period of time (a) has first been treated by Tsui and Chang (1990, 1992). They model the problem as a quadratic assignment problem, which minimizes the total distance traveled by the shipments within the cross dock and conduct computational experiments based on a representative distribution of shipments among related sites.

Gue (1999) develops a rough model of the freight flows in a cross docking center that applies a so-called look-ahead-policy. In such a framework the cross docking center usually monitors the queue of inbound trucks and assigns outbound trucks to empty outbound gates, before the inbound truck arrives at an inbound gate, to minimize workers travel distances. The author proposes a linear program (LP) determining a typical freight mix received at an inbound gate. This model aims, in line with the look-ahead-policy, at minimizing the workers travel distances and is employed repeatedly in order to evaluate different layouts for cross docking centers by comparing the total minimum labor costs of the layouts. The layout decision, here, is which gate serves as an inbound gate and which one serves as an outbound gate. In a case study, the author shows that applying a look-ahead policy leads to 15–20 percent lower labor costs, compared to a first-come-first-served-policy. Finally, using an optimal layout results in further labor cost savings of up to 30 percent.

Bartholdi and Gue (2000) further refine the models proposed by Gue (1999). Several important characteristics are added to the cost model such as interference among forklifts, dragline congestion, and congested floor space. Layout optimization is conducted with the help of a simulated annealing algorithm. Each candidate layout is evaluated by the associated labor cost. In a case study, the authors optimize the layout of the outbound gates of a cross docking facility and record the productivity of the facility the month before and three months after the implementation. The overall labor productivity increases by 11.7 percent, even though the layout of the inbound gates remains unchanged.

If we have much more destinations to serve than dock doors available, we rather have a short-term truck scheduling problem deciding on a variable assignment of destinations (b). Among these approaches, transshipment times inside the terminal are either (i) completely neglected, modeled by a (ii) constant time premium, or (iii) by a transshipment time that depends on the distances between the respective dock pairs.

The first stream of literature (i) assumes that freight can move through the terminal in zero time. Chen and Lee (2007) draw on the similarities between assigning inbound and outbound trucks to two dedicated gates and flow shop problems. They consider two sets of jobs, one for the inbound gate and one for the outbound gate. Jobs can be assigned to the second machine, representing outbound trucks being assigned to the outbound gate, only after a subset of jobs on the first machine, representing inbound trucks being unloaded, is completed. To solve the problem an approximation algorithm and a branch-and-bound algorithm are developed and evaluated. Chen and Song (2009) generalize the problem to parallel machines on both stages representing multiple inbound gates and multiple outbound gates. The problem is shown to be strongly NP-hard. The authors developed a mixed integer program (MIP) for the described problem and employed a standard solver. For larger instances they propose four different heuristics. Finally, Cota et al. (2016) build on the results of Chen and Song (2009) and propose a new model formulation and a polynomial time approximation algorithm.

The second stream of literature (ii) assumes that freight flows through the terminals in a given constant time which is independent of the gates. Often, this assumption is justified since only a single inbound gate and a single outbound gate is considered. Yu and Egbelu (2006) determine the freight assignment to outbound trucks and the docking sequences of inbound and outbound trucks simultaneously. The described cross docking facility hereby consists of a single inbound gate and a single outbound gate and an area for temporary storage. The objective of the work is the minimization of the total operation time of a cross docking facility where inbound trucks arrive at the facility and are unloaded without preemption, pallets are shipped through the facility on (one or multiple) conveyor belts and ultimately loaded onto outbound trucks leaving the facility. Furthermore, the authors assume a constant changeover time for trucks. To determine the optimum docking sequences and product assignments an MIP is developed and several greedy heuristics are proposed.

Boysen, Fliedner, and Scholl (2010) investigate the assignment and scheduling problem of trucks at two dedicated gates. Inbound trucks deliver known quantities of different pallets which can intermediately be stored. Outbound trucks pick up known quantities of different pallets and can be processed only if these pallets are available in the storage. With respect to the freight flow through the cross dock the authors assume that the travel time is a known constant and equal among all pallets. The problem is proven to be NP-hard and several lower bounds are provided. Furthermore, they describe a decomposition strategy, which optimizes one side of the cross docking facility, i.e., with inbound or outbound, while fixing the other. Arabani, Ghomi, and Zandieh (2009) consider a similar cross dock setting but have multiple objectives. They aim at just-in-time processing of trucks and a minimization of the makespan. They propose three different meta-heuristic approaches: Genetic algorithm (GA), particle swarm optimization and differential evolution. Boysen (2010) revisits the assignment and scheduling problem defined by Boysen et al. (2010) and extends it to multiple gates on both sides but does not allow any temporary storage of pallets. Different objectives are considered, i.e., minimizing the total flow time, total processing time or total tardiness. The authors provide a binary linear program (BIP) and exact as well as heuristic solution methods.

Alpan, Larbi, and Penz (2011) allow storage of pallets at the terminal and additionally consider the option to let outbound trucks approach the terminal multiple times in order to successively relieve the respective gate. The objective is to minimize the operational costs of the cross docking facility. These costs are calculated by adding up temporary storage costs and replacement costs for outbound trucks. The authors solve the problem to optimality with a dynamic programming (DP) approach where possible and apply a bounded dynamic programming approach for larger instances.

Finally, in the third stream of literature (iii) travel times of freight are positive and depend on the specific inbound gate and the outbound gate. Van Belle, Valckenaers, Berghe, and Cattrysse (2013) present a model formulation for the truck assignment and scheduling at multiple inbound gates and multiple outbound gates. Two objectives are considered, namely minimization of total travel time of pallets through the cross dock and minimization of total tardiness of trucks. Travel times of pallets are assumed to depend on the distance between inbound gate and outbound gate. The authors develop a MIP model and solve the problem using standard solvers and a tabu-search approach. Assadi and Bagheri (2016a,b) build up on the problem proposed by Arabani et al. (2009) and develop a model where trucks arrive over time aiming at minimum total earliness and tardiness costs. Given the complexity of this problem, they succeed in solving the problem using standard solvers only for very small instances. In order to tackle larger instances, two different meta-heuristics are developed and applied. In Boysen, Briskorn, and Tschöke (2013a), the authors investigate a problem first introduced in Boysen and Fliedner (2010), which treats the truck assignment and scheduling for the case of fixed departures of outbound trucks. In addition to a model formulation and complexity observations, they develop several approaches that decompose the problem. They investigate first-sequence-then-assign (FS) and first-assign-then-sequence (FA) procedures and benchmark them with a basic simulated annealing algorithm. The results of a computational study show that assigning trucks to door first leads to very promising solutions.

5. ASS in the postal service industry

Within the past decades, significant growth in the postal service (also denoted as CEP, i.e., courier, express and parcel) industry forced carriers to explore new technologies and logistic patterns (Ducret, 2014), which resulted in the construction of large package sorting facilities (e.g. Airportzentrale, 2014). Sorting facilities of the postal service industry are similar to cross docking terminals, but decision makers face additional challenges, i.e., huge volumes of shipments and great time pressure. The daily delivery volume of UPS, for instance, amounts to 18.3 million packages and documents, while serving more than 220 countries and territories worldwide (UPS, 2017). Due to the ambition of more and more carriers to deliver letters and parcels within the next or even same day of ordering (McAree, Bodin, & Ball, 2002; Werners & Wülfing, 2010), time windows for the consolidation process tighten even more. Depending on the function of a hub, the means of transportation serviced, typically, vary. Truck-based terminals are usually applied for consolidation within continental transport (Werners & Wülfing, 2010), whereas facilities at airports serve as

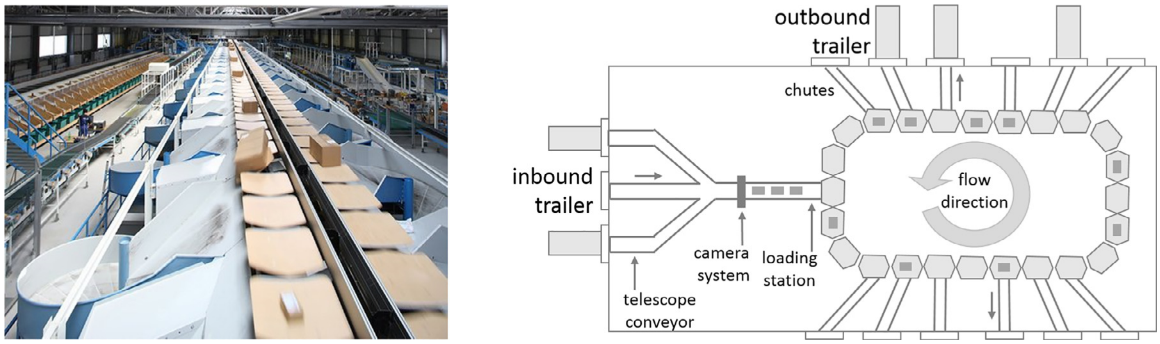

Fig. 5. Crisplant LS-4000E™ tilt tray sorter (Source: Beumer Group) (left) and terminal of the postal service industry (right).

intermodal hubs and handle their goods via aircraft (Boysen & Fliedner, 2011). In the latter case, inbound and outbound vehicles are small towing vehicles that connect aircraft parked on the apron with the terminal building. A tilt tray conveyor, which is often applied in the postal service industry, and a schematic terminal layout is depicted in Fig. 5.

Once inbound trailers arrive at dock doors, shipments are unloaded by logistics workers onto telescope conveyors (see Fig. 3 (right)), which move shipments towards camera systems identifying the items (Billo, Porter, Mazumdar, & Brown, 2003). Afterwards, shipments follow a path to the main conveyor, which is connected by a *single* or *multiple* inbound stations. Carriers such as DHL, Hermes, or UPS apply either *loop* systems or even larger *bi-directional* conveyor systems (Briskorn et al., 2017). The parcel hub of Germany's second largest postal service provider Hermes in Friedewald (Germany), for example, operates with two loop sorters of 700 m length flowing in the same direction and installed on top of each other, that are served via a single inbound station (Boysen et al., 2017b). The DHL sorter at the hub in Leipzig is 6,500 m long and has four loading stations and two parallel loop conveyors that travel in opposite directions (Fedtke & Boysen, 2017b). Other terminals, such as the UPS airhub in Cologne (Germany), apply even more complex ASS consisting of a *network of conveyors* (see also McWilliams et al., 2005). Many of these more complex systems are subdivided into a primary sorter, which directs shipments towards multiple alternative secondary sorters (Jarrah et al., 2014). These two-stage systems avoid that each shipment has to travel in a single loop through the complete terminal building so that, especially in a vast airhub terminal, transportation time on the conveyor is reduced. Once a shipment on the main sorter reaches its dedicated outbound chute, it is directed into its outbound station, which is typically another telescope conveyor directly leading into the trailer servicing the respective destination (Fedtke & Boysen, 2017b). Less frequented outbound destinations are also collected in roll containers, which are rolled towards trailers once the loading time is reached (Werners & Wülfing, 2010; Zenker & Boysen, 2018). At the outbound side, most terminal managers aim at a *fixed* assignment of outbound destinations (Fedtke & Boysen, 2017b) so that the door assignment remains stable over a longer period of time. However, there also exist terminals applying a *variable* assignment that varies over the day (Jarrah et al., 2014).

In the following section, we survey the existing literature on layout planning, inbound and outbound planning as well as short-term scheduling.

5.1. Layout planning

When designing the layout of ASS in terminals of the postal service industry simulation plays an important role in order to quantify the impact of different layout alternatives on the operational sortation performance. Consequently, we, first, discuss two papers, which focus solely on how to design suitable simulation models of ASS in postal industries.

Geinzer and Meszaros (1990) investigate how a conveyor, as a part of a larger distribution system, can be modeled and how the different modeling options affect the resulting simulation in terms of model size and complexity, computer memory requirements, simulation run time and accuracy. Different degrees of aggregation are considered when building the simulation model. The authors apply those modeling options to a Federal Express facility, where all conveyors handle high volumes and have a single induction and outbound point. The authors report that balancing simulation accuracy and simulation effort is a delicate matter. After choosing an appropriate model it has been implemented and successfully used by the project engineers to improve the flow within the facility and to better react to occurring conveyor failures.

Bard (1997) conducts a broad comparison of different available software packages and tests the applicability of such in the field of postal industry ASS management. In detail, the author checks all 172 software packages listed by Rodrigues (1994) with regard to their eligibility for modeling a complex postal industry distribution center. Based on very general requirements, the author only finds three simulation software packages to be capable to appropriately model such a complex system, namely Arena, ProModel and Witness. To evaluate all three candidates, the author implements a simulation of a facility of the Canada Post Corporation with all three software packages and measures the performance of each of them against benchmark criteria. After developing a scoring system together with the Canada Post Corporation, ProModel proved to be the superior software package for that specific case.

Specific layout aspects are treated by the following papers. Myers et al. (2003) assist the sorting system manufacturer Lockheed Martin Distribution Technologies (LMDT) in finding an optimal layout for a mail-sorting system for the US Postal Service. The authors implement simulation models of possible predefined layouts, with Arena, even before the physical prototyping. To formalize the requirements imposed by users and their accomplishment, the authors initially translate the requirements into numerical values and derive confidence intervals in order to measure the precision of their simulation. This idea is an adaption of the comparison concept of standards and test systems proposed by Kelton and Law (2000). Finally, the authors come to the conclusion that a pre-prototyping assessment of different designs can be a powerful strategy to quickly exclude non-suited layouts.

Fedtke and Boysen (2017b) systematically investigate different design alternatives of closed-loop sortation systems in parcel distribution centers. By studying several real-world closed-loop parcel sortation systems, the authors identify four layout parameters that differ among existing systems, namely the number of induction points, the number of parallel recirculation conveyors, their flow direction, and the number of trays. An extensive simulation study finally shows the impact of the different layout options. While the number of trays in the system does not affect the throughput, increasing the number of induction points improves throughput significantly. Furthermore, increasing the number of parallel recirculation conveyors is shown to generally improve the throughput. This effect is even more significant if they rotate in opposite direction.

5.2. Inbound problems

At the inbound side of a facility of the postal service industry, the flow of shipments onto the sorter can be manipulated (i.e., with regard to the assignment of shipments to inbound stations and the sequence per inbound station). Both decisions are directly influenced by the processing schedules of the vehicles delivering the shipments. In truck-based terminals, the assignment of trucks to inbound docks and the sequence of processing per dock is to be planned. This problem is also known as the truck scheduling problem (Boysen & Fliedner, 2010). We, however, focus on those papers where the impact of the truck schedule on the ASS is (at least indirectly) considered. Analogously, in an airhub incoming flows arriving on aircraft and (after unloading) on towing vehicles need to be scheduled. Existing approaches in both directions are surveyed in the following.

McWilliams et al. (2005) present an inbound truck scheduling problem common to freight consolidation terminals and, in particular, in hubs of the postal industry. Inbound trucks are to be assigned to multiple inbound gates and, furthermore, unloading sequences of inbound trucks assigned to the same inbound gate have to be determined. The objective is to minimize the total sortation time while balancing the workload in different parts of the system. The ASS is assumed to be a network of loop conveyors. Each inbound gate is connected with one of several primary sorters, while each secondary sorter is connected with each primary sorter. Trucks deliver identical amounts of items and are assumed to be available at the beginning of the planning horizon. The authors present a lower bound and a simulation-based scheduling algorithm (SBSA). The SBSA is basically a GA in which unloading schedules constitute the members of the population and each member's fitness is determined by a simulation model. The computational results show that the proposed approach reduces the total sortation time significantly compared to real-world practices, but computation times are extremely high. McWilliams (2005) presents an alternative to the above mentioned SBSA for the same problem. The new algorithm proceeds in three stages. A simulated annealing algorithm is used in order to find good permutation lists of inbound trucks. The objective value associated with each list is determined in the remaining two stages. First, each list is interpreted as unload schedules using a list scheduling algorithm, which assigns one inbound truck after the other to the first available inbound gate. Second, the simulation presented in McWilliams et al. (2005) is used to evaluate the unload schedule. The solution quality is similar to the one in McWilliams et al. (2005), but the computational effort is slightly reduced. McWilliams (2009b) and McWilliams and McBride (2012) present a GA and a beam search algorithm solving the problem stated in McWilliams et al. (2005) and prove that the beam search algorithm is superior to all the earlier presented solution methods for large instances, which are especially relevant for practical cases in major sortation facilities of the leading postal service providers. McWilliams (2010a) considers a generalization of the problem presented in McWilliams et al. (2005) by no longer assuming equal sized inbound trucks. In addition to developing a MIP model representing the problem, the author develops another GA based on the one presented in McWilliams (2009b). A computational study proves that the newly established algorithm outperforms all of the previously mentioned ones, but still requires quite long computational times. Finally, McWilliams (2010b) compares different solution approaches for the problem defined in McWilliams et al. (2005).

McWilliams (2009a) develops a solution procedure for a variant of the problem presented in McWilliams et al. (2005). A dynamic programming (DP) approach explicitly aims at balancing the workload within the sortation facility and at its outbound gates. The proposed procedure can be interpreted as an automated dispatcher, who chooses a queuing inbound truck whenever an inbound gate becomes idle and assigns it to the free gate. The choice hereby is taken in a way that minimizes imbalances inside the sortation system. A computational study compares this DP approach with the methods proposed in McWilliams et al. (2005) and McWilliams (2005) and comes to the conclusion, that the automatic dispatcher approach achieves much better results in a similar computational time. Haneyah, Schutten, & Fikse (2013) extend the problem definition of McWilliams (2009a) by considering travel times of shipments within the sortation system. Furthermore, they consider differently prioritized outbound trucks and allow them to be delayed. The authors adapt the algorithm presented in McWilliams (2009a) to consider both extensions and show that their newly developed algorithm is nearly exactly as good and quick as the more restrictive version of McWilliams (2009a).

The study of Boysen et al. (2017b) reveals that truck schedules can directly affect the performance of an ASS, because they determine the sequence in which shipments flow into the system. When assigning inbound trucks to docks such that many shipments reach their dedicated outbound gates prior to reaching the next inbound station, multiple tray loads per cycle of the loop conveyor are enabled. They formalize the resulting truck scheduling problem, prove computational complexity and provide suited solution procedures based on interval scheduling. Their results show that the throughput of the ASS can considerably be increased with on optimized inbound schedule of delivery trucks.

Much less research is dedicated to airhubs. The inbound problem of an airport terminal of the postal service industry is merely investigated by Boysen and Fliedner (2011). The idea behind the study is to optimize the aircraft landing schedules in a way that the number of landed passengers, the number of landings per airline, and the number of passengers per airline are leveled. Although an ASS is not explicitly modeled, the determined schedules lead to a steady inbound stream of shipments into the sorter and, thus, help to avoid peaks loads. The authors develop suited decomposition approaches and dynamic programming based procedures for each objective.

5.3. Outbound problems

On the outbound side of a terminal, outbound stations of the ASS need to be assigned to outgoing destinations. Typically, postal service providers aim at fixed assignments that remain stable for a few months (Boysen & Fliedner, 2010). However, if the number of outbound destinations to be serviced exceeds the number of available docks, a sharing of docks is inevitable. A shared assignment can either be realized on a fixed or variable basis. The former means that two or more outbound destinations share a dock over a longer period of time so that their loads line up in a staging area in front of the dock and the pallets or roll containers are loaded once the respective trailer is docked (Zenker & Boysen, 2018). A variable assignment may change from day to day and depends on the short-term availability of docks. Dock changes may require a relocation of shipments that have arrived from the sorter in the meantime (Jarrah et al., 2014). In the following, we review the existing literature on this topic.

Masel and Goldsmith (1997) present a simulation study to systematically assign shipments leaving the main conveyor to so-called sort positions on a fixed basis. Here, a sort position is an accumulation lane leaving the central recirculation conveyor. Multiple destinations may use the same sort position and one destination may require multiple sort positions depending on the volumes of the destinations. The parcel sortation facility under consideration applied simple rule-based approaches for assigning sort positions to arriving shipments. The authors simulate the facility, which has 110 sort positions for 14,000 destinations and include potential defects of conveyor components. They implement the model using AutoMod based on a given CAD layout file in order to examine the resulting package flows and sortation times of the system when applying different assignment rules. Using the simulation model, the destination assignment can be chosen among given alternatives minimizing the total sortation time without overloading system areas.

Werners and Wülfing (2010) propose a three-dimensional assignment problem generalizing the setting considered in Masel and Goldsmith (1997). The three dimensions correspond to arriving shipments, the sort positions and the delivery tours, while the assignment of shipments to inbound gates and the assignment of outbound gates to sort positions are assumed to be given and fixed by the strategic management. The authors aim at minimization of total transportation effort (given by the distance and the amount of shipments between two gates), that suits the need of a parcel sortation facility of Deutsche World Post Net (DPWN) best. The authors model the problem as an MIP and decompose the model hierarchically into a three-level problem. To account for stochasticity, a robust optimization model is developed. Finally, the results of the computational study and a comparison to the current operations within the concerned DPWN facility show that both, the robust and the deterministic model solutions, achieve significant improvements compared to the status quo.

Zenker and Boysen (2018) explore the problem which outbound destinations should share a dock, if not enough docking space is available to process each destination separately. They aim to minimize the maximum inventory accumulating in the staging areas of a facility with a recirculating tilt tray conveyor with multiple outbound stations. After formulating a mathematical model, the authors propose a two stage solution procedure. At first, they determine a feasible start solution, that is iteratively improved by a greedy, tabu search, fix and optimize, or combined solution procedure. Within a computational study and a sensitivity analysis, they investigate the impact of several problem parameters, e.g., the dock/truck ratio and the level and variation of processing times and accumulation factors.

The destination-loader-door-assignment problem is studied by Jarrah et al. (2014). Their first objective is to assign destinations to consecutive doors so that the number of changes of destination-to-door assignments from one sort to the next is minimized. The second and third objectives are to minimize the number of workers at the doors and to evenly distribute the volume of packages assigned to each worker. The considered sorting process consists of two stages. At first, a primary sorter is loaded via multiple inbound stations and shipments are spread among several secondary sorters each equipped with multiple outbound stations. They propose a mathematical model that minimizes the weighted sum of switches and workers and develop a solution procedure that exploits the hierarchical nature of the problem. They furthermore apply their approach to a case study with four hubs in North America.

Fedtke and Boysen (2017b) propose a model formulation for the so-called destination assignment problem with multiple line segments which can be described as follows. A closed-loop conveyor layout with multiple conveyor segments, each consisting of an induction point and a given number of outbound points, is considered. Shipments with individual destinations arriving at inbound gates are given. It is to be decided which inbound gate uses which induction point to feed the sorter. Furthermore, destinations are to be assigned to outbound stations. Both decisions imply the path of each shipment through the sorter. The objective is to minimize total travel-time of shipments on the ASS. The authors consider variants with a single closed-loop conveyor and multiple parallel bi-directional conveyors. They prove NP-hardness of both variants and evaluate different heuristics.

5.4. Short-term scheduling

Once the layout of the sorter and all inbound and outbound destinations are fixed, there are still some short-term decisions that have to be taken. Specifically, there may be flexibility with regard to the access time and the exact path of each shipment through the ASS. Literature on these aspects is, however, rare.

Briskorn et al. (2017) address the short-term scheduling and routing of shipments in a loop conveyor. They define eight different problem settings that vary in three aspects: First, the induction point is predetermined for each shipment or still to be decided. Second, the sequence of shipments approaching inbound station is predetermined for each shipment or still to be decided. Third, the recirculation conveyor consists of a single loop or of two loops rotating in opposite directions. The authors present an exact DP approach and a heuristic for each of the eight variations and present an extensive computational study, which shows that a systematic optimization can result in a significantly higher throughput.

Different loading rules for a tilt tray conveyor are evaluated in the study of Fedtke and Boysen (2017b). They evaluate the first-empty-tray, the shortest-distance, and the minimum-tilt-time rule both for a single loop and a bi-directional system with multiple inbound station and derive some advise under which circumstances what rule is preferable.

6. ASS for baggage handling

In addition to passenger transport, the handling of baggage is a fundamental task at airports and embraces the transport of baggage towards outgoing and back from incoming aircraft. Before loading an aircraft, different inbound streams have to be merged and synchronized. Baggage from check-in counters, incoming flights, other means of transportation (e.g., from trains delivering early baggage checked in directly in the city center, which is, for instance, possible in Vienna) and storage areas for early check-in might be moved by internal devices towards the sortation system, are fed into the system, and are merged and distributed towards the respective outbound station for a departing aircraft (Abdelghany et al., 2006). On the other side, baggage from incoming aircraft has to be transported towards the correct baggage claim (Tošić, 1992). Tight flight schedules only leave small time windows between arrival and departure of baggage, nonetheless reliability is crucial for customer satisfaction (see Section 1). Furthermore, the airline industry has some of the highest rates of work-related back and shoulder injuries in all of the private sector (Korkmaz et al., 2006), due to the manual transfer of heavy and bulky baggage. Therefore, ASS became a basic element for rapid, fully-automated handling of huge amounts of shipments in modern airports. Fig. 6 (left) schematically depicts the baggage flow at airport terminals.

On the inbound side, an ASS receives baggage from different areas of the airport terminal. Check-in baggage is either directly transported to the baggage handling system (BHS) or temporarily stored for later flights. Moreover, the inbound baggage from incoming flights enters the sorting system as well. Some travel agencies offer to ship baggage via several means of transportation (e.g.,

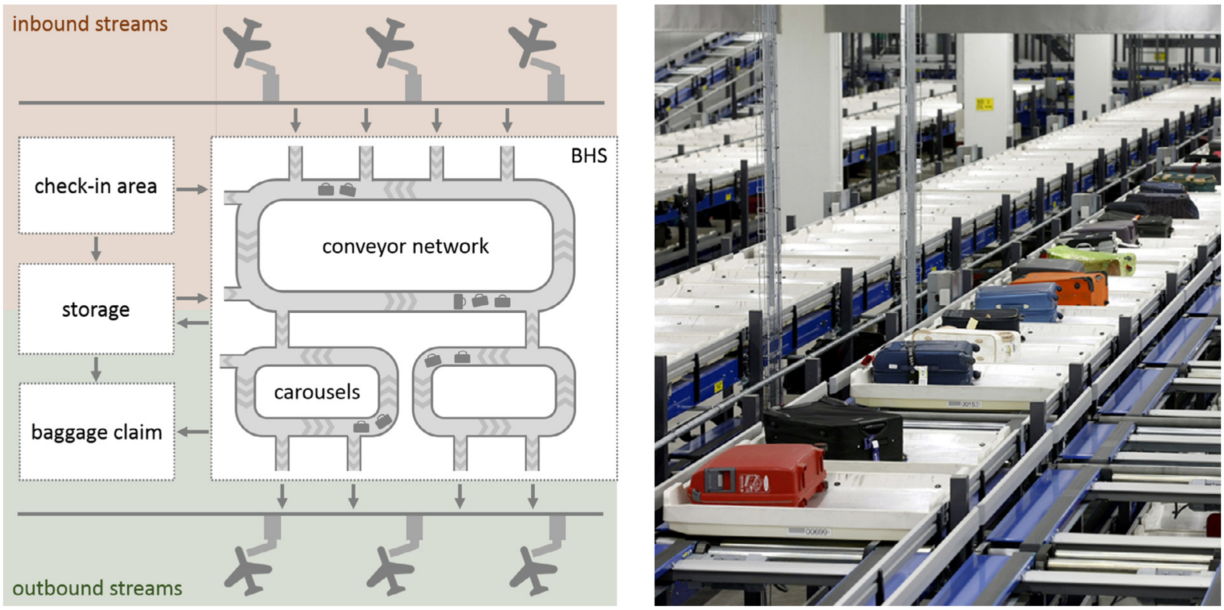

Fig. 6. Typical setup of a baggage handling systems at an airport terminal (left) and the CrisBag® tote system belt conveyor (Source: Beumer Group) (right).

trains) directly to the travel destination. These streams of incoming baggage are also directed to the BHS. Thus, we typically have *multiple* inbound stations with baggage. The ASS itself, typically, consist of a *complex conveyor networks* including several junctions, straight and recirculating elements even connecting different terminals, and baggage carousels (Khosravi, Nahavandi, & Creighton, 2009). The applied technologies for BHS are manifold. Besides classic belts and tilt tray conveyors, conveyors with special boxes (see Fig. 6 (right)), each holding a single bag, exist. Although not being part of this survey, also destination-coded vehicles (DCVs) are applied (see Tarău et al., 2011) here. These individual carts propelled by linear induction motors are mounted on tracks and travel with varying velocity and diverging distance among each other. On the outbound side of the ASS, claim baggage or terminating baggage has to be sorted towards the correct baggage claim carousel and outbound baggage towards designated outbound stations for outgoing flights or other forms of transport. This also includes transfer baggage, i.e., baggage from incoming flights to outgoing flights. Typically, there are far more destinations of incoming and outgoing flights than outbound stations so that a *variable* assignment is required.

Before we summarize previous research on optimizing baggage handling in airports, we start with some preliminary remarks why the importance of baggage handling varies with the specific baggage stream to be handled (see also Tošić, 1992). For outgoing flights, the check-in counters close long before take-off and, thus, typically allow enough time to transport the baggage to the planes. Analogously for incoming international flights, passenger travel times to the baggage claims, including immigration and security checks, are usually long enough to accept non-optimal travel times for baggage. Passengers arriving at an airport via national flights reach the baggage claims much earlier, because immigration is not required. For these passengers, a quick baggage handling reduces their waiting times for their bags and, thus, influences customer satisfaction. Of utmost importance, however, is the quick and reliable handling of transfer baggage. If connecting flights are missed, passengers reach their destinations without their baggage, which greatly influences customer satisfaction (U.S. Department of Transportation, 2009) and requires costly additional deliveries.

In spite of this importance, baggage handling is often set aside in favor reducing walking distances of transfer passengers in the scientific literature. Airports are generally assessed and globally ranked on various factors. One of the driving factors in most rankings is the “ease of transit”, which is often measured in passengers’ walking distances. Thus, most transit airports focus their efforts on minimizing the walking distances of passengers to get higher ranking and allow shorter time windows for transits. Consequently, the assignment of incoming and outgoing flights to gates, for instance, typically aims at minimizing passengers’ walking distances rather than optimizing baggage handling processes explicitly (see Bouras, Ghaleb, Suryahatmaja, & Salem, 2014). However, if the walking distance between the gates is minimizes, implicitly the distance for rapid baggage transfers among aircraft tends to be reduced too.

As a result of this, we found no scientific publications dealing with an optimization of layout decisions related to baggage handling systems explicitly. However, in order to evaluate the performance of existing ASS or specific layouts, a vast amount of simulation studies has been conducted. We, therefore, dedicate the first subsection to this topic. Subsequently, we survey literature on inbound, outbound, and short-term scheduling problems related to BHS.

6.1. Layout planning

To cope with the complex structure of BHS and their stochastic influences, simulation models provide an appropriate instrument for investigations. By simulating the entire system (or single elements) the baggage flows can be analyzed, bottlenecks can be identified, and conclusions with regard to design and layout aspects can be drawn. Thus, again, simulation has significant meaning for decision support related to ASS. The application of BHS simulations dates back to the late 1960s. In a first basic simulation study, Robinson (1969) concludes that the baggage sorter performs better for business flights than for vacation flight, as the density of bags per passenger is relatively low and fewer bottlenecks occur.

Specific BHS setups of real-world airports are investigated in (Jaroenpuntaruk & Miller, 1995; Savrasovs, Medvedev, & Sincova, 2009). Jaroenpuntaruk and Miller (1995) use the general purpose simulation language GPSS/H to simulate the baggage facility at O’Hare International Airport in Chicago. They conduct experiments in order to study the impact of different the baggage transfer volumes on the system behavior. The BHS at Riga airport is simulated by Savrasovs et al. (2009). The discrete-event simulation (implemented with ExtendSim) includes several check-in counters, an equipment area, a security check area, as well as conveyor networks connecting the segments. The authors study the system's performance for different rates of passenger flow and concluded, that the bottleneck appears between the check-ins and the equipment area.

Hanta and Pozivil (2010) simulate a baggage handling system using Witness PwE, a discrete event simulation software. The BHS is modeled as a sortation system with several inbound gates for transfer, terminating, and check-in baggage. Baggage passes through a two-stage sortation process. The primary sortation process diverts transfer baggage and check-in baggage to conveyor belts leading to outgoing flights. Terminating baggage is forwarded to a single recirculation conveyor, which assigns the baggage to the designated baggage claim carousel in the terminal. Their simulation model proves that very complex airport baggage handling systems can be modeled and used to evaluate different possible changes in the system structure. Liu and Liu (2014) propose an alternative simulation model, using AutoMod. The system consists of two induction lanes that transport baggage to the main sortation system. The main sortation system, in this case, is a recirculation conveyor with tilt trays, which automatically ejects the baggage into one of multiple outbound chutes. Another simulation of Aguilera-Venegas, Galán-García, Mérida-Casermeiro, and Rodríguez-Cielos (2014) includes several inbound belts from check-in counters a circular conveyor system, several outbound carousels, as well as storage and security areas. With the help of their model, the performance of specific setups is measured and a stress test for different numbers of active check-in counters is conducted.

6.2. Inbound problems

Typically, BHS receive bags from multiple inbound stations, e.g., check-in counters, storage areas or arriving aircraft. The inflow from these sources into the ASS can mainly be influenced by determining the assignment of flights to gates, the assignment of baggage to recirculation conveyors and the balancing of workload among check-in counters.

Hu and Di Paolo (2007) consider a variant of the aircraft gate assignment problem (for a general survey on this problem see Dorndorf, Drexl, Nikulin, & Pesch, 2007), which treats, among other goals, the optimization of the baggage handling operations. The problem is to assign all incoming and outgoing flights to the gates of the airport, such that multiple objectives are optimized. Specifically, they aim to minimize the passengers' walking distances between gates, waiting times of aircraft, and travel distances of baggage between gates. The authors model the problem as a MIP and solve it using a GA.

Frey, Artigues, Kolisch, and Lopez (2010) study the assignment and scheduling of transfer baggage arriving on incoming flights with given arrival times to one of multiple recirculation conveyors. These conveyors are employed to sort the transfer baggage for outgoing flights. For each incoming flight an arrival process is given which represents the amount of arriving baggage over time while outgoing flights have a fixed time for take off. The sortation system consists of several recirculation conveyors and a central capacitated storage area where bags are intermediately stored. On the recirculation conveyor, baggage passes several stations where it can be loaded on carts servicing outgoing flights. The problem is modeled as a variant of the resource-constrained project scheduling problem and proven to be NP-hard. Analogously, Barth, Holm, and Larsen (2013) consider the transfer baggage problem, which is to assign arriving baggage from incoming flights to an inbound area of the BHS. Their system consists of several induction lanes with specific capacities for baggage. The main conveyor system transports the baggage to handling facilities of the outgoing flights where they are containerized and transported to outgoing flights. The ASS is not explored in detail, but transportation times from the inbound areas to the corresponding handling facilities are considered. The problem is modeled as an MIP model with multiple objectives, minimizing the baggage transfer times and the number of baggage items missing their flights while balancing the workload at the inbound areas. The resulting model is solved with CPLEX and successfully evaluated with data from Frankfurt Airport.

Kim, Kim, and Chae (2017) treat the balancing of baggage handling performance of a check-in area shared by multiple airlines. The considered system at Gimhae International Airport consists of several check-in counters connected to a single belt conveyor which directs baggage through an inspection area onto the main conveyor. They develop a dynamic re-allocation algorithm of check-in counters to airlines and their current flights. A simulation study shows that this approach reduces the risk of blockages on the inbound conveyor and decreases passenger waiting.

6.3. Outbound problems

Analogously to the inbound side, BHS have multiple outbound stations. Either, bags leaving the ASS have to be directed to claim carousels where passengers can pick up their bags or to outbound stations dedicated to leaving aircraft. In both cases, there are far more destinations than outbound stations so that a variable assignment over time is required. Other than in the previous applications of ASS, when assigning the baggage from incoming flights to baggage claims it is possible that multiple flights share an outbound station as long as the total capacity is not violated. Thus, slightly different assignment decisions arise compared to the previous areas.

Barth (2013) studies the baggage carousel assignment problem, which is to assign one or more baggage claim carousels in the arrival hall to incoming flights. Each carousel has a given capacity for terminating baggage and feasible assignments are precalculated. The objective is to minimize operational costs composed of direct costs and costs arising from violations of soft constraints. Direct costs consist of the passengers' walking distances and the travel distances of the baggage. Soft constraints aim to avoid the assignments of two large aircraft without sufficient time gap to the same carousel or unbalanced workloads among baggage claims. While the ASS is not considered in detail, the precalculated feasible assignments and the various operational costs factors implicitly take the ASS into account. The problem is formulated as a MIP model solving a set partitioning problem and evaluated with real-world instances from Frankfurt Airport. An analogous problem is studied by Frey, Kiermaier, and Kolisch (2017a) for the Franz Josef Strauss Airport in Munich. Here, baggage claims are either fed by directly connected infeed stations for baggage carts or from the BHS, which collects baggage from multiple remote infeed stations. They minimize the claim carousels' utilization and, additionally, the passengers' walking distances and waiting times. As a result peaks at the baggage carousels are reduced by 38% and passenger waiting by 11%.

The planning of outbound baggage and its assignment to outbound stations where baggage carts are filled servicing the planes is investigated by Frey, Kolisch, and Artigues (2017b). In order to avoid disruptions of the system they aim to minimize workload peaks over the entire system. To do so, they determine the assignment flights to outbound stations but also the scheduling of each flight's baggage handling, i.e., the starting times of the sorting processes. The resulting mathematical model is solved by a column generation approach. The computational results applying real-world data from Munich airport indicates a performance increase up to 60% compared to manual solutions. A similar problem is treated by Huang, Mital, Goetschalckx, and Wu (2016). They determine the time windows for the baggage handling processes at different outbound stations. These time windows, called activities, depend on stochastic influences, i.e., the number of bags for each flight and potential delays of aircraft. They model the problem as a stochastic vector assignment Problem and apply the progressive hedging method to solve it. A case study with data from a major airport in Asia shows, that the current manual solution of human decision makers are clearly outperformed. Abdelghany et al. (2006) call their problem variant the pier assignment problem where, however, again the assignment of outbound flight to outbound stations over time is determined. In a multi-objective approach, they aim at sufficient time buffers between two successive flights assigned the same station and short baggage transportation distances. Furthermore, they aim to assign two flights having the same destination to the same station so that late baggage missing their actual flight is directly at the right position where also the later flight with the same destination is handled. They propose a simple greedy heuristic for solving the problem and test it with data from O'Hare international airport. Even more objectives are considered by Ascó et al. (2014). They also include robustness issues to avoid negative impact of delayed flights, a fair workload distributions among outbound stations, and deviations from preferred stations. They test different heuristic solution procedures with lockahead and improvement for solving the resulting multi-objective problem. Computational tests with data from London Heathrow airport show that selecting the right optimization procedure can greatly improve the solution quality.

6.4. Short-term scheduling

After fixing the general flow of baggage from inbound to outbound stations, the exact path of baggage though the ASS has to be determined. Mainly, this involves decisions on control and merging rules for bags when changing between different sorter segments and conveyor belts.

Lin, Shih, Huang, and Chiu (2015b) simulate the baggage handling system of Taiwan Taoyuan International Airport using the System Modeling Language (SysML). The baggage handling system consists of multiple inbound areas for check-in baggage and arriving baggage and carousels, each with conveyors transporting the baggage to a network of sorters leading to unload stations or buffer zones. Every buffer zone has the same capacity and baggage sizes are assumed to be identical. After validation of the proposed model, different control rules are studied, such as minimizing the total travel time between inbound areas and unload stations or buffer zones. In a follow-up paper Lin, Liou, and Chiu (2015a) extend their simulation of control rules. This study shows that the overall operation time of the system can be reduced by 23% during peak-time, if appropriate routing policies are selected. Johnstone, Creighton, and Nahavandi (2015) specifically focus the merging bags from two conveyors onto one receiving belt. They compare the application of fixed belt segments and compare them with variable segments that vary with bag sizes.

7. Summary and future research needs

The detailed breakdown of all 70 reviewed papers and how they fit into our framework is presented in the appendix. As a summary, we present the stacked bar chart of Fig. 7 where we see the number of papers published in our four areas of ASS application over time. Except for warehousing, where ASS have a long lasting tradition and the number of published papers remains rather stable, the other three areas of application gain increasing attention over the years. In spite of the tremendous scientific work that already exists on ASS, we see a wide range of future research

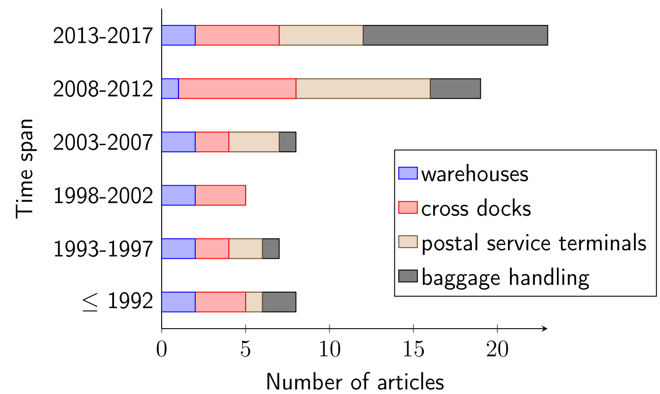

Fig. 7. Statistics to the number of reviewed articles.

challenges which we elaborate in the following. We structure these challenges according to our framework which differentiates four areas of application and four decision areas (see Section 2).

- Warehousing: One major problem of ASS in warehouses applied to sort customer orders picked under a batching and/or zoning policy is, that they are hardly scalable. Many warehouses face workloads that extraordinarily vary both during the week (i.e., most online retailers have their peak load after the weekend) and over the year (e.g., due to cyber monday or end of season sales). Moreover, many online retailers have to adopt their order fulfillment capacities to their growth rates so that they aim to flexibly adapt their sortation capacity. ASS, however, require plenty fixedly installed hardware, which is hardly adaptable over a shorter period of time. Due to this drawback, many online retailers apply manual sortation processes where they apply so-called put walls (see Boysen et al., 2018). A put wall is a simple reach-trough rack separated into multiple shelves each temporarily dedicated to a specific customer order. On the other side of the wall reside packing workers, who empty shelves and pack completed orders into cardboard boxes. By adapting the workforce either to put items into the shelves or to pack them on the other side, sortation capacity can flexibly be altered. Future research should integrate the scalability aspect when evaluating the basic trade-off between the higher investment costs for ASS and higher personnel costs of manual sorting processes. Also, ideas how to make ASS better scalable seem very welcome.
- Cross docks: Many cross docking terminals do not apply a sorter at all. Goods are moved with forklifts or pallet jacks inside the terminal instead. Future research should investigate the general trade-off between higher investment costs if an ASS is installed and larger personnel costs if goods are sorted manually in more detail. Decision support under which circumstances either solution is preferable seems highly welcome.
- Postal service industry: Two general ASS layouts dominate within the postal service industry. Terminals either consist of two-stage sorting systems where a pre-sorter feeds multiple main sorters (Jarrah et al., 2014) or they provide a single loop (or a bi-directional system where conveyors are erected on top of each other) (Briskorn et al., 2017; Fedtke & Boysen, 2017b). The former provides redundancy and the system can still process even if one of the main sorters suffers a breakdown or is maintained. The reduced structure of the latter layout eases control and avoids scheduling and workforce decisions with regard to equalizing the workload among multiple main sorters. Systematic decision support with regard to which layout is best suited for what specific situation, however, is yet missing.

- Airport: Baggage handling systems (BHS) in airports consist of complex conveyor networks, whose structure is strongly influenced by the terminal layout. Therefore, existing research mainly treats isolated case studies, whose results are barely generalizable. A valid general research question, however, is the comparison between the conveyor-based ASS surveyed in this paper and destination-coded vehicles (DCVs). The latter are shuttle-based system transferring bags individually (see Tarău et al., 2011) so that they promise more flexibility and, thus, a higher throughput. However, quantifying the potential gains of DCV in dependency of different structural properties of the BHS (and comparing these gains with the higher investment costs) would be valuable decision support for airport managers having to decide on the proper type of BHS.
- Layout: Scalability is promised by an alternative sorting system, which is not based on fixed pathways. In this system (see Section 2), individual autonomous robots move along a grid of squares on the shop floor towards one of many scuttles dedicated to the shipments's destination. A first application of such a grid-based sorting system consisting of a fleet of Hikvision robots is reported for a large warehouse in Hangzhou (China) (Tompkins International, 2017; People's daily, 2017). Each robot having the size of an autonomous vacuum cleaner is able to tilt its tray so that the item on top slides into the scuttle where items are collected for a specific destination. Such a system is easily scalable by adding or removing robots. However, the layout and operational decision problems to be solved for such a problem are completely different to those of traditional ASS. Another novel system applied in warehouses, e.g., the Zalando distribution center in Mönchengladbach (Germany) (RP Online, 2016), is a bag sorter. They adapt the basic principle of hanging goods handling and move items in small bags hanging from trolley conveyors. After picking, individual items are filled each in a hanging bag, which are automatically moved into an intermediate storage area. Upon request, bags are moved out of their storage lanes, automatically sorted into the right sequence via a system of switches and conveyed to packing stations. Here, items arrive in the right sequence so that one order after the other can be packed into its dedicated cardboard box. The big advantage of a bag sorter is that multiple orders can queue in front of their destination so that gridlocks due to unavailable packing can only occur if all queues are full. Scientific research on these systems is (to the best of the authors' knowledge) yet not available.
- Inbound side: Existing research mainly focuses the assignment of incoming shipments to inbound stations (if multiple alternatives exist). In some applications, however, either a manual pre-sorting process or an intermediate transfer into and out off some automated storage and retrieval system (e.g., a crane-operated high-bay rack, see Boysen & Stephan, 2016) also allows for a manipulation of the sequence of incoming items. The first paper investigating the impact of the inbound sequence on sortation performance is the one of Briskorn et al. (2017). More research into this direction seems valuable.
- Outbound side: An operational problem on the outbound side, which (to the best of the authors' knowledge) has not received any attention in the literature yet, is staffing and workforce scheduling in the outbound area. In warehouses, for instance, the accumulation lanes of an ASS are, typically, equipped with beacons that indicate completed orders. The packers filling these orders into cardboard boxes can either be assigned a fixed number of (typically consecutive) lanes or can freely roam through the packing area. The former is inflexible and may lead to additional idle time of packers, but has the advantage that errors can definitely be assigned to its culprit. Comparing both assignment policies and deriving suited scheduling procedures for both of them has not yet been considered in the literature. This is especially unsatisfying, because inefficient packing processes may lead to blocked outbound stations, which in the worst case may lead to gridlocks (see Section 3.2) on the main conveyor. Analogously, the staffing and workforce scheduling problems in cross docking terminals and facilities of the postal service industry have not yet been considered. Here, the workforce loading sorted shipments into the trailers needs to be scheduled. Each outbound station can be assigned its separate logistics worker. This rules out congested outbound stations and, thus, gridlocks, but comes for the price of high personnel costs. Instead, most terminals assign each worker multiple outbound stations. Decision support how many stations each worker should support is yet missing. This, also, holds true for the question when each worker should switch among stations in his/her area. A decision rule for this task has to consider the distributions of parcel arrivals, the fill levels of lanes (and the remaining time up to a congestion), and the worker's walking times among lanes.
- Scheduling: Especially in the fixed-path ASS surveyed in this paper, there is often not much flexibility when feeding shipments onto the main sorter and channeling them towards outbound stations. Therefore, short-term scheduling may not be the foremost lever for increasing system performance. However, there are still quite a few interesting short-term scheduling problems whose impact on system performance should be evaluated. For instance, the application of in-floor chain conveyors in cross docking terminals gives rise to a novel scheduling problem. If a single logistics worker is assigned the task to load a trailer with the dedicated pallets circulating on a dragline at given positions, this problem resembles single machine scheduling where moving a pallet into a trailer constitutes a job. The main peculiarity, however, is that the processing times for a job and the sequence-dependent setup times among jobs (i.e., walking back from the trailer to the current position of the next pallet) vary with the circulation of the conveyor. Time-dependent processing times are a well researched topic in machine scheduling (e.g., see the survey paper of Cheng, Ding, & Lin, 2004), but circulating items constitute a specific metric for the processing times, which has not been considered yet.

Generally, it can be stated that there is a need for holistic models that consider the mutual interdependence between inbound processes, main conveyor, and outbound processes in more detail. Without shipments arriving from the inbound side, the main sorter runs idle. On the other hand, if the throughput of the main sorter increases, more shipments can be loaded. Analogously, congestion in outbound stations impacts the throughput of the main sorter. The feedback links between all three basic elements should be modeled in more detail so that additional knowledge on the dynamics of ASS can be gained.

Finally, transferring the knowledge gained for ASS to related settings and vice versa seems promising. For instance, AGVs transferring containers between quay cranes processing container vessels and gantry cranes servicing container yards can be interpreted as some kind of ASS (see Section 2). Recall that these systems have been excluded from our survey, because AGVs travel individually along their paths (and not in a steady flow). However, transferring the knowledge between both fields may help to derive future port systems that rather resemble a traditional ASS or more flexible ASS based on individual flows may be possible. Analogously, knowledge transfers with individual destination-coded vehicles for baggage sorting and container shuttles in rail-rail transshipment yards (see Fedtke & Boysen, 2017a) seem desirable. The same holds true for a knowledge transfer among the four applications surveyed in this paper and their specific ASS implementations.

In light of these manifold future research tasks it can be projected that ASS will remain a fruitful field of research in the foreseeable future.

Appendix. Summary of reviewed articles

Table A.1 summarizes the content of the reviewed articles with regard to the area of application, the considered sorting system, the treated problem and the solution method. Note, that many authors propose a MIP formulation before introducing a solution method. However, 'MIP formulation' is only stated, if no other solution approach was developed. Within the table, we make use of the following abbreviations:

Application: **W:** warehouse, **C:** cross dock, **P:** postal service industry, **A:** airport
Inbound type: **single:** single inbound station, **multi:** multiple inbound station, **nem:** not explicitly modeled
Main conveyor: **li:** linear sorter, **loop:** loop sorter, **bi:** bi-directional, **net:** conveyor network, **nem:** not explicitly modeled
Outbound type: **fixed:** fixed assignment of outbound destination, **variable:** variable assignment, **nem:** not explicitly modeled
Problem type: **layout:** layout problem, **in:** inbound problem, **out:** outbound problem, **short:** short-term scheduling problem
Solution method: **sim:** simulation study or simulation-based solution method, **DP:** dynamic programming-based approach or beam search, **SA:** simulated annealing, **GA:** genetic algorithm, **PS:** particle swarm algorithm, **DEC:** decomposition approach, **PR:** priority rule, **B&B:** branch-and-bound, **PTAA:** polynomial time approximation algorithm, **DLB:** dynamic load balancing, **GRASP:** greedy randomized adaptive search procedure, **LS:** local search, **CG:** column generation, **SVAP:** stochastic vector assignment problem, **PHM:** progressive hedging method, **MNFM:** maximum network flow model, **TS:** tabu search, **F&O:** fix-and-optimize, **FIFO:** first-in-first-out

Table A1
Summary of reviewed articles.

Ref.	App.	Inbound	Conveyor	Outbound	Problem	Approach
(Aguilera-Venegas et al., 2014)	A	multi	net	variable	layout	sim
(Alpan et al., 2011)	C	multi	nem	variable	in, out	(bounded) DP
(Al-Sultan & Bozer, 1998)	C	nem	nem	nem	layout	SA
(Arabani et al., 2009)	C	single	nem	variable	in, out	GA, PS, differential evolution
(Assadi & Bagheri, 2016b)	C	multi	nem	variable	in, out	SA, GA
(Assadi & Bagheri, 2016a)	C	multi	nem	variable	in, out	differential evolution, SA
(Bakst et al., 1996)	C	nem	nem	nem	layout	sim
(Bard, 1997)	P	nem	nem	nem	layout	simulation software comparison
(Barth, 2013)	A	multi	net	variable	out	MIP formulation
(Barth et al., 2013)	A	multi	nem	fixed	in	MIP formulation
(Bartholdi & Gue, 2000)	C	multi	nem	fixed	layout, in, out	SA
(Boysen, 2010)	C	multi	nem	variable	in, out	DEC, DP
(Boysen et al., 2013a)	C	multi	nem	variable	in	DEC, SA
(Boysen et al., 2016)	W	single	li	variable	in	PR-based heuristic, DP, sim
(Boysen et al., 2017b)	P	multi	loop	fixed	in	DEC, DP, sim
(Boysen & Fliedner, 2011)	P	single	nem	nem	in	DP, DEC
(Boysen et al., 2010)	C	single	nem	variable	in, out	DEC
(Bozer & Sharp, 1989)	W	single	li, loop	variable	layout	sim
(Bozer et al., 1988)	W	single	loop	variable	out, short	sim
(Briskorn et al., 2017)	P	multi	loop, bi	fixed	short	DP, DEC
(Chen & Lee, 2007)	C	single	nem	variable	in, out	approximation algo, B&B
(Chen & Song, 2009)	C	multi	nem	variable	in, out	Johnson's rule-based heuristics
(Cota et al., 2016)	C	multi	nem	variable	in, out	PTAA
(Fedtke & Boysen, 2017b)	P	multi	loop, bi	fixed	layout, in, out, short	sim, SA
(Frey et al., 2010)	A	multi	net	variable	in	DEC
(Frey et al., 2017a)	A	multi	net	variable	out	hybrid heuristic (GRASP, LS)
(Frey et al., 2017b)	A	multi	net	variable	out	guided CG approach
(Gallien & Weber, 2010)	W	single	loop	variable	in	sim
(Geinzer & Meszaros, 1990)	P	nem	nem	nem	layout	sim. modelling option eval.
(Gue, 1999)	C	multi	nem	fixed	in, out	LP based local search
(Haneyah et al., 2012)	P	multi	net	fixed	in	DLB algorithm
(Hanta & Pozivil, 2010)	A	multi	net	variable	layout	sim
(Hu & Di Paolo, 2007)	A	multi	nem	variable	in, out	GA
(Huang et al., 2016)	A	multi	net	variable	out	SVAP, PHM
(Jaroenpuntaruk & Miller, 1995)	A	multi	net	variable	layout	sim
(Jarrah et al., 2014)	P	multi	net	nem	out	B&B
(Johnson, 1998)	W	single	loop	variable	out, short	analytical model
(Johnson & Lofgren, 1994)	W	single	loop	fixed	layout	sim
(Johnstone et al., 2015)	A	multi	li	single	short	FIFO-based algorithms
(Kim et al., 2017)	A	multi	loop	nem	in	re-allocation algorithm, sim
(Kizilaslan et al., 2103)	W	single	loop	single	in	analytical model
(Lin et al., 2015a)	A	multi	net	variable	short	sim
(Lin et al., 2015b)	A	multi	net	variable	short	sim
(Liu & Liu, 2014)	A	multi	loop	variable	layout	sim
(Masel & Goldsmith, 1997)	P	multi	loop	fixed	out	sim

(continued on next page)

Table A1 (*continued*)

Ref.	App.	Inbound	Conveyor	Outbound	Problem	Approach
(McWilliams, 2005)	P	multi	net	fixed	in	sim, SA
(McWilliams, 2009a)	P	multi	net	fixed	in	DLB algorithm
(McWilliams, 2009b)	P	multi	net	fixed	in	GA
(McWilliams, 2010b)	P	multi	net	fixed	in	local search, SA
(McWilliams, 2010a)	P	multi	net	fixed	in	GA
(McWilliams et al., 2005)	P	multi	net	fixed	in	sim, GA
(McWilliams & McBride, 2012)	P	multi	net	fixed	in	BS
(Meller, 1997)	W	single	loop	variable	out, short	DEC
(Myers et al., 2003)	P	nem	nem	nem	layout	sim
(Owyong & Yih, 2006)	W	multi	nem	nem	in	improvement algorithm, sim
(Pan et al., 2008)	C	nem	nem	nem	layout	multi-commodity MNFM
(Petersen, 2000)	W	nem	nem	nem	layout	sim
(Robinson, 1969)	A	multi	net	variable	layout	sim
(Rohrer, 1995)	C	nem	nem	nem	layout	sim
(Russell & Meller, 2003)	W	nem	nem	nem	layout	sim
(Savrasovs et al., 2009)	A	multi	net	variable	sim	sim
(Sharp & Liu, 1990)	C	nem	loop	nem	layout	analytical model, ILP
(Tošić, 1992)	A	nem	nem	nem	nem	general review
(Tsui & Chang, 1990)	C	multi	nem	fixed	in, out	LS
(Tsui & Chang, 1992)	C	multi	nem	fixed	in, out	B&B
(Van Belle et al., 2009)	C	nem	nem	nem	layout	holonic multi-agent approach
(Van Belle et al., 2013)	C	multi	nem	variable	in, out	TS
(Werners & Wülfing, 2010)	P	multi	net	variable	out	DEC
(Yu & Egbelu, 2006)	C	single	nem	variable	in, out	greedy heuristics
(Zenker & Boysen, 2018)	P	single	loop	nem	out	greedy heuristic, TS, F&O

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